

BOOKBINDERS CLUB

Predictive Targeting & Campaign Strategy

A direct-mail book retailer deciding who should receive a catalog for a new title — using customer purchase history to separate likely responders from the rest of the file.

This case applies Enginius predictive-modeling case data to a Product Marketing Manager portfolio exercise. Campaign cost and margin figures used in the targeting economics are illustrative, not Bookbinders' actual financials (see Limitations).

The Business Problem

Bookbinders Club reaches its customers entirely through mailed catalogs. Every new title raises the same question before a dollar is spent.

- 1 The default assumption: mail the whole file and let response rate fall where it may. Historically, only 25% of customers respond to any given offer — three in four mailings produce no sale.
- 2 Customers aren't interchangeable. Purchase history shows very different category interests — children's, cooking, DIY, art — so a title like an art history book won't resonate with everyone the same way.

OBJECTIVE

Determine which customers are most likely to respond to a new title — and use that to decide who gets mailed, who doesn't, and how the campaign budget should be allocated.

Response probability is used to rank and tier customers; it is a targeting input, not a guarantee of purchase (see Limitations).

Data Overview

1,600

Customers analyzed
(calibration sample)

25%

Historical response rate
(400 of 1,600 customers)

11

Columns in source
dataset

10

Predictors modeled

Target variable

Choice (0/1) — responded to the mailed offer

Predictors

Gender · Amount purchased · Frequency · Last purchase · First purchase · P_Child · P_Youth · P_Cook · P_DIY · P_Art

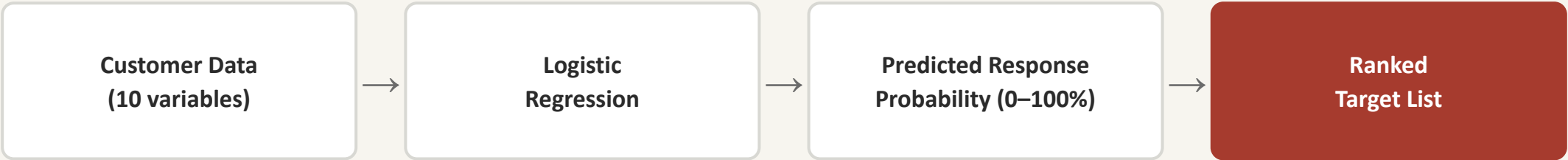
Method

Logistic regression (binary response model, estimated on the full calibration sample)

Frequency table

Alternative 0 (no purchase): 1,200 customers (75.0%) · Alternative 1 (purchase): 400 customers (25.0%)

Methodology: From Purchase History to a Response Score



The model learns, from 1,600 past responses, how strongly each customer characteristic relates to buying — then scores every customer from 0% to 100% likely to respond.

MODEL FIT SUMMARY

80.1%

Overall classification accuracy

40.3%

Actual responders correctly flagged

93.4%

Actual non-responders correctly flagged

The model reliably separates high- and low-probability customers, even though it doesn't call every individual case correctly — that's what targeting uses, not raw accuracy.

Model Fit

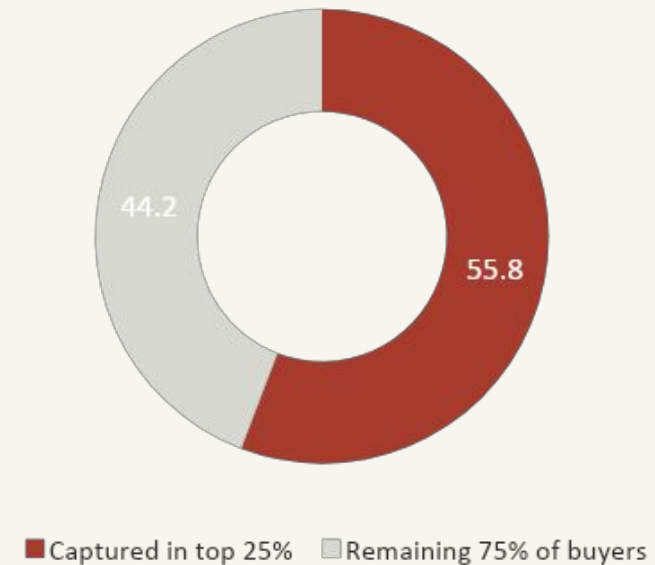
80.1%

Overall classification accuracy (in-sample)

3.0×

Lift over random targeting at the top 10% of the ranked list

Buyers captured at the top-25% cutoff



44% of likely responders still sit outside the top quartile of the ranked list — lift is strong, not perfect (see Limitations).

Key Drivers: What Moves Response

P_Art (art-book history)

STRONGEST POSITIVE

Coefficient	+0.6736
z-value	5.29
p-value	< 0.0001

P_Cook (cookbook history)

STRONGEST NEGATIVE

Coefficient	-0.9295
z-value	-7.79
p-value	< 0.0001

Category-specific interest is the strongest force in the model — a customer's history with art books helps, while cooking, DIY, and children's-book history works against this specific offer.

Reading the Model Correctly: Nuances & Cautions

First Purchase (tenure)

NOT SIGNIFICANT

Coefficient: -0.0148 p-value: 0.2486

How long someone's been a customer doesn't predict this offer's response once other factors are controlled.

Frequency (total past purchases)

SIGNIFICANT — NEGATIVE

Coefficient: -0.0756 p-value: < 0.0001

More purchases overall doesn't mean more interest in this book — being active isn't the same as being interested.

Last Purchase (recency)

COUNTER-INTUITIVE

Coefficient: $+0.6211$ p-value: < 0.0001

Customers who purchased longer ago were, if anything, more likely to respond — worth validating before acting on it.

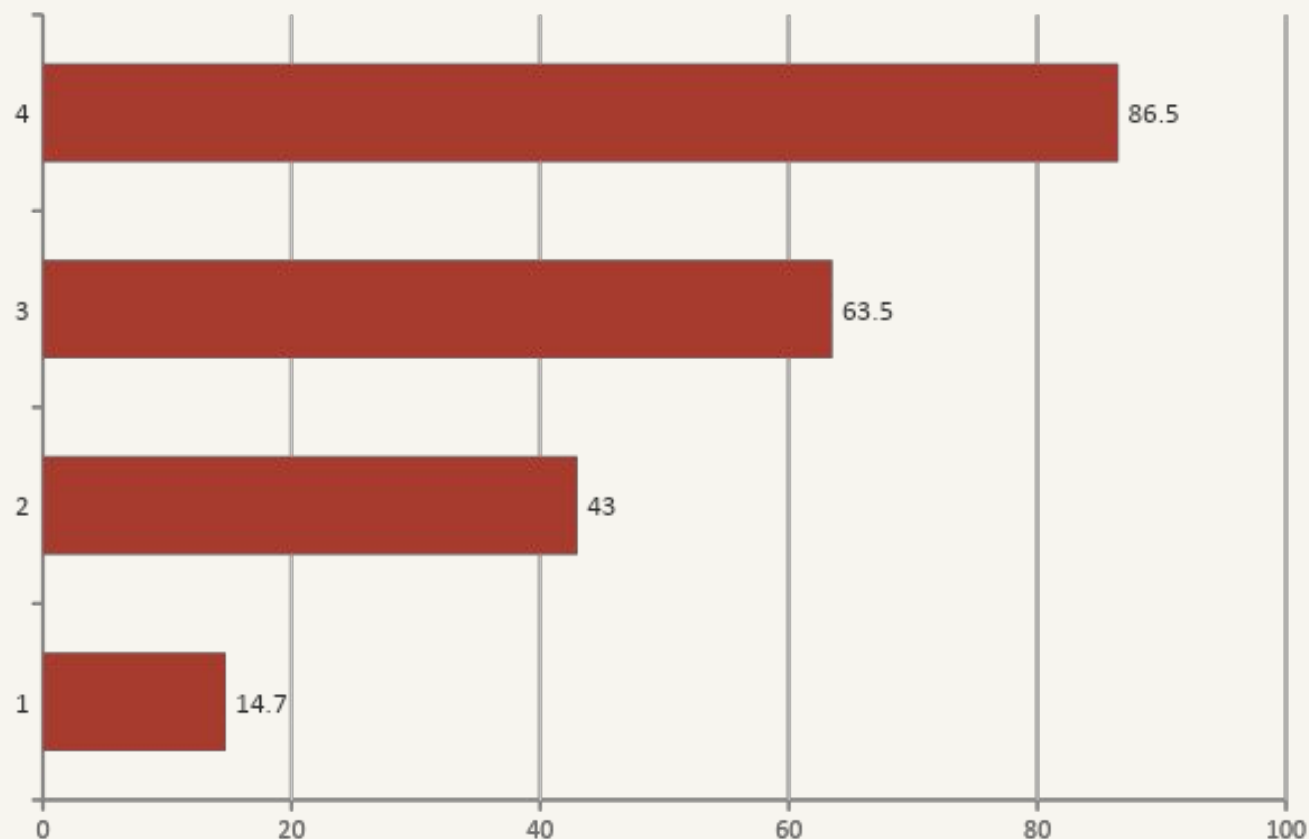
Gender

NEEDS VERIFICATION

Coefficient: -0.8643 p-value: < 0.0001

A real effect, but the exported data doesn't say which coded value is which — confirmed before use as a lever.

Targeting Tiers: Who Responds



70+ pts

Spread from the least- to most-likely tier — larger than the effect of any single predictor alone.

Highest:

Top 0–5% of file — 86.5% predicted response

Lowest:

Bottom 75% of file — 14.7% predicted response

→ *see targeting economics, next*

Category Affinity: What Predicts Interest

Elasticities show how sensitive response probability is to a small change in each predictor — how much the model's story depends on that variable.

Pulls toward response

P_Art (art-book history)	+20.0%
Amount purchased (lifetime spend)	+21.9%

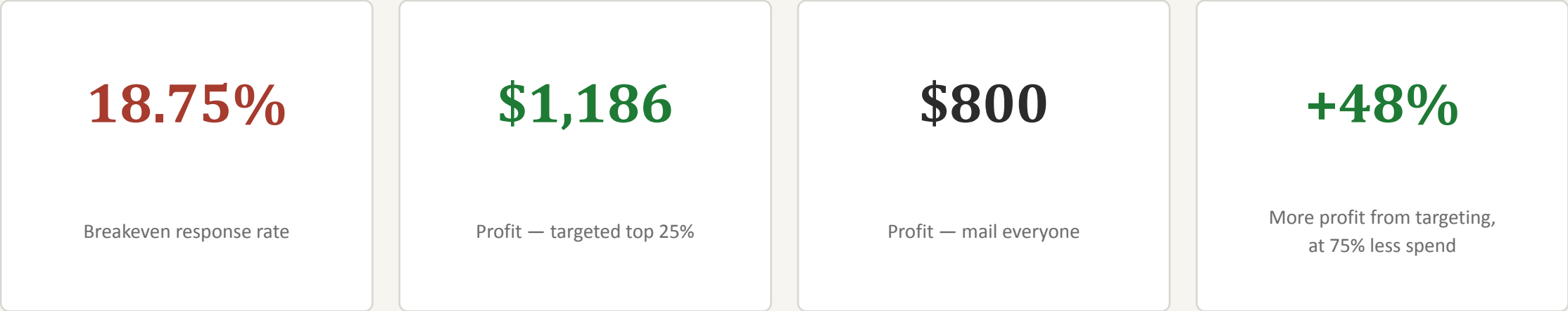
Pulls away from response

Frequency (total past purchases)	−42.5%
P_Cook (cookbook history)	−35.0%
P_Child (children's-book history)	−30.8%
P_DIY (DIY-book history)	−17.9%
P_Youth (youth-book history)	−12.2%

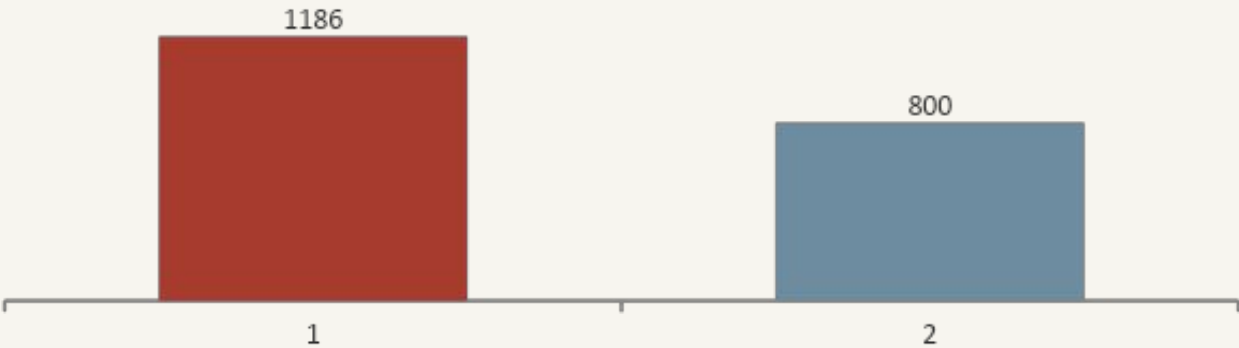
A clean pattern: general purchase activity and category interest pull in opposite directions. The most active customers overall aren't automatically the right audience for this title — what they buy matters more than how much.

Targeting Economics*

**Illustrative campaign assumptions — \$1.50 cost per mailed catalog, \$8.00 contribution margin per book sold — used to demonstrate the framework (see Limitations).*



Mailing the excluded 75% of the file would add an estimated \$1,800 in postage against \$1,414 in expected revenue — a projected loss of about \$386 on that portion alone.



Limitations & Data Constraints

80.1% in-sample accuracy leaves meaningful room for error at the individual customer level.

- 1 Model performance (80% accuracy, 3.0× lift) was measured in-sample — on the same data used to fit the model — not validated on a fresh holdout set.
- 2 Mailing cost (\$1.50) and contribution margin (\$8.00) are illustrative assumptions, not Bookbinders' actual campaign financials.
- 3 The exported model summary doesn't specify which coded value of “Gender” maps to which real-world label — excluded from the final recommendation.
- 4 The “Last Purchase” (recency) finding runs counter to typical intuition and should be validated on new data before being treated as a rule.
- 5 The source PDF is a generic Enginius predictive-modeling export; business framing (title, catalog context) was applied for this case exercise.
- 6 Response probability estimates individual likelihood, not certainty — some customers outside the recommended tier will still respond, and some inside it won't.

Recommendation: A Tiered Mailing Strategy

Author's recommendation — a decision framework, not a guarantee of the illustrative dollar figures

TARGET

Mail the top 25% of the file, ranked by predicted response probability (~400 of 1,600 customers) — split into a high-touch top-10% tier and a standard 10–25% tier.

EXCLUDE

Skip the bottom 75% for this print offer; route that segment to a lower-cost channel (e.g. email) instead of dropping the relationship entirely.

WHY

Under illustrative economics, targeting the top 25% is projected to produce ~48% more profit than mailing everyone, while spending 75% less on printing and postage.

VALIDATE

Before committing the full budget: confirm real cost/margin figures, validate the model on a holdout sample, and pilot the tiered plan on one region first.

Measurement Roadmap & Next Steps

- 01** Validate the model on a fresh holdout sample before the cutoff drives real mail spend.
- 02** Confirm actual mailing cost and per-book margin, and re-run the breakeven cutoff with real figures.
- 03** Resolve the Gender coding against the source data dictionary before treating it as a targeting lever.
- 04** Track actual vs. predicted response after the mailing, and use it to recalibrate the model for the next title launch.

Questions & Discussion